/* ACADEMY SOFTWARE FOUNDATION

Overview

March 2026

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Antitrust Policy Notice

Linux Foundation meetings involve participation by industry competitors, and it is the intention of the Linux Foundation to conduct all of its activities in accordance with applicable antitrust and competition laws. It is therefore extremely important that attendees adhere to meeting agendas, and be aware of, and not participate in, any activities that are prohibited under applicable US state, federal or foreign antitrust and competition laws.

Examples of types of actions that are prohibited at Linux Foundation meetings and in connection with Linux Foundation activities are described in the Linux Foundation Antitrust Policy available at linuxfoundation.org/antitrust-policy. If you have questions about these matters, please contact your company counsel, or if you are a member of the Linux Foundation, feel free to contact Andrew Updegrove of the firm of Gesmer Updegrove LLP, which provides legal counsel to the Linux Foundation.

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Why Open Source?

Why Open Source?

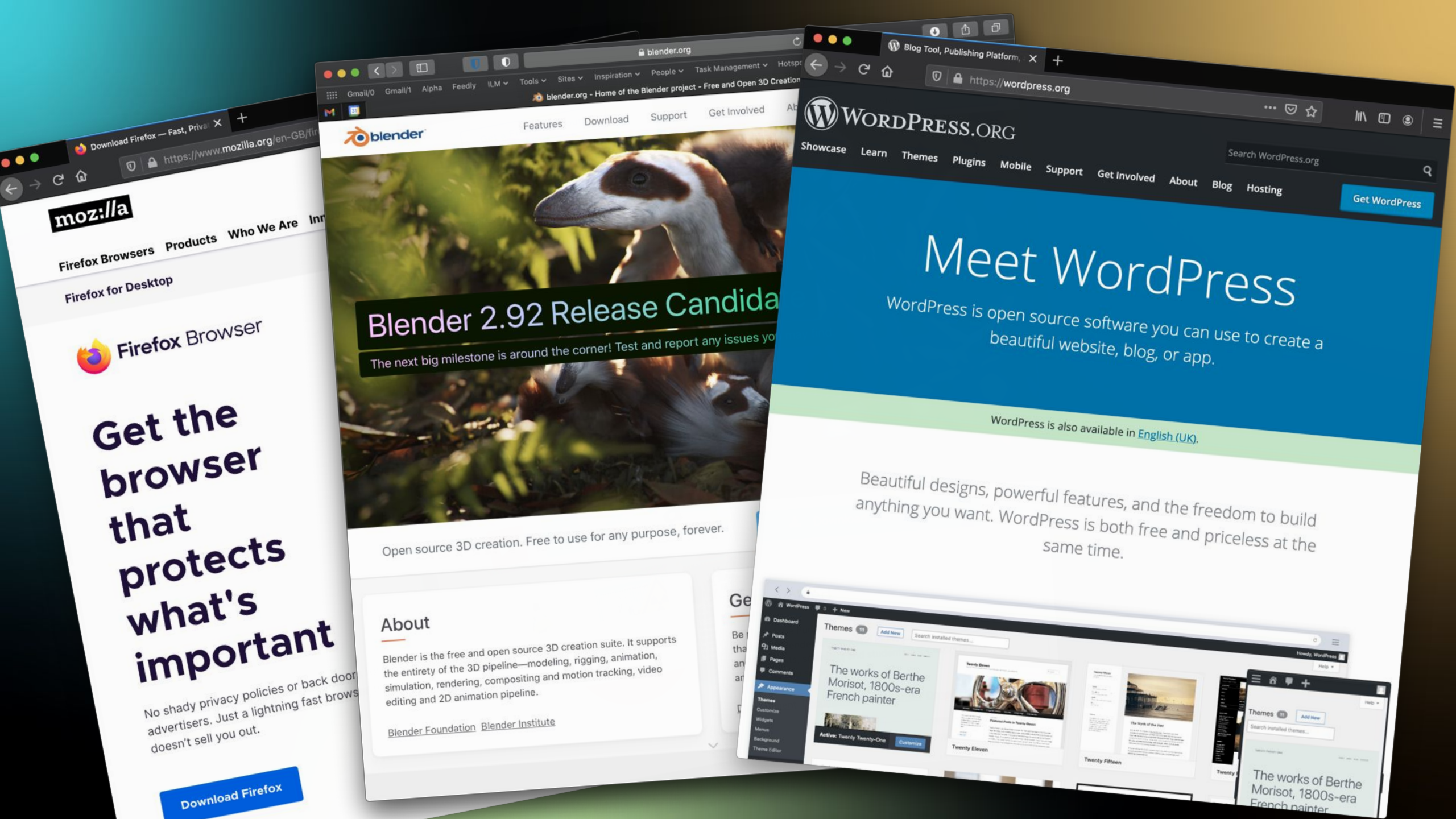
- Software development has a **cost**
  - Development
  - Maintenance
  - Documentation
  - Training
- Open Source ecosystems **share** those costs
- Empower developers to lead development

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Open Source Projects




Download Firefox — Fast, Priva
https://www.mozilla.org/en-GB/fir
moz://a
Firefox Browsers
Products
Who We Are
Firefox for Desktop
Firefox Browser
Get the browser that protects what's important
No shady privacy policies or back door
advertisers. Just a lightning fast brows
doesn't sell you out.
Download Firefox
blender.org
Gmail/0
Gmail/1
Alpha
Feedly
ILM
Tools
Sites
Inspiration
People
Task Management
blender.org - Home of the Blender project - Free and Open 3D Creation
blender
Features
Download
Support
Get Involved
Blender 2.92 Release Candida
The next big milestone is around the corner! Test and report any issues yo
Open source 3D creation. Free to use for any purpose, forever.
About
Blender is the free and open source 3D creation suite. It supports the entirety of the 3D pipeline—modeling, rigging, animation, simulation, rendering, compositing and motion tracking, video editing and 2D animation pipeline.
Blender Foundation
Blender Institute
Blog Tool, Publishing Platform
https://wordpress.org
WordPress.org
Showcase
Learn
Themes
Plugins
Mobile
Support
Get Involved
About
Blog
Hosting
Search WordPress.org
Get WordPress
Meet WordPress
WordPress is open source software you can use to create a beautiful website, blog, or app.
WordPress is also available in English (UK).
Beautiful designs, powerful features, and the freedom to build anything you want. WordPress is both free and priceless at the same time.
Themes
Add New
The works of Berthe Morisot, 1800s-era French painter
Active: Twenty Twenty-One
Customize
Twenty Eleven
Twenty Fifteen

Open Source: Innovation Accelerator

Standardization on open technologies	Neutral environment for collaboration & cross-pollination	Innovation multiplier	Enables better interoperability	Minimizes fragmentation
Qualifies reference architectures	Lowers barriers to enter a new domain	Enables new business opportunities	Leads to better products	Allows shared cost of development

About the Academy Software Foundation

Open Source in Entertainment

How the Academy Software Foundation Creates Shared Value

A report from

THE LINUX FOUNDATION | Research

Written by
Barbara Robertson

report.aswf.io



DUNE *Image Courtesy of DNEG © Legendary Warner Bros. Entertainment Inc. All Rights Reserved.*

An Open Source Foundation Developed in Partnership



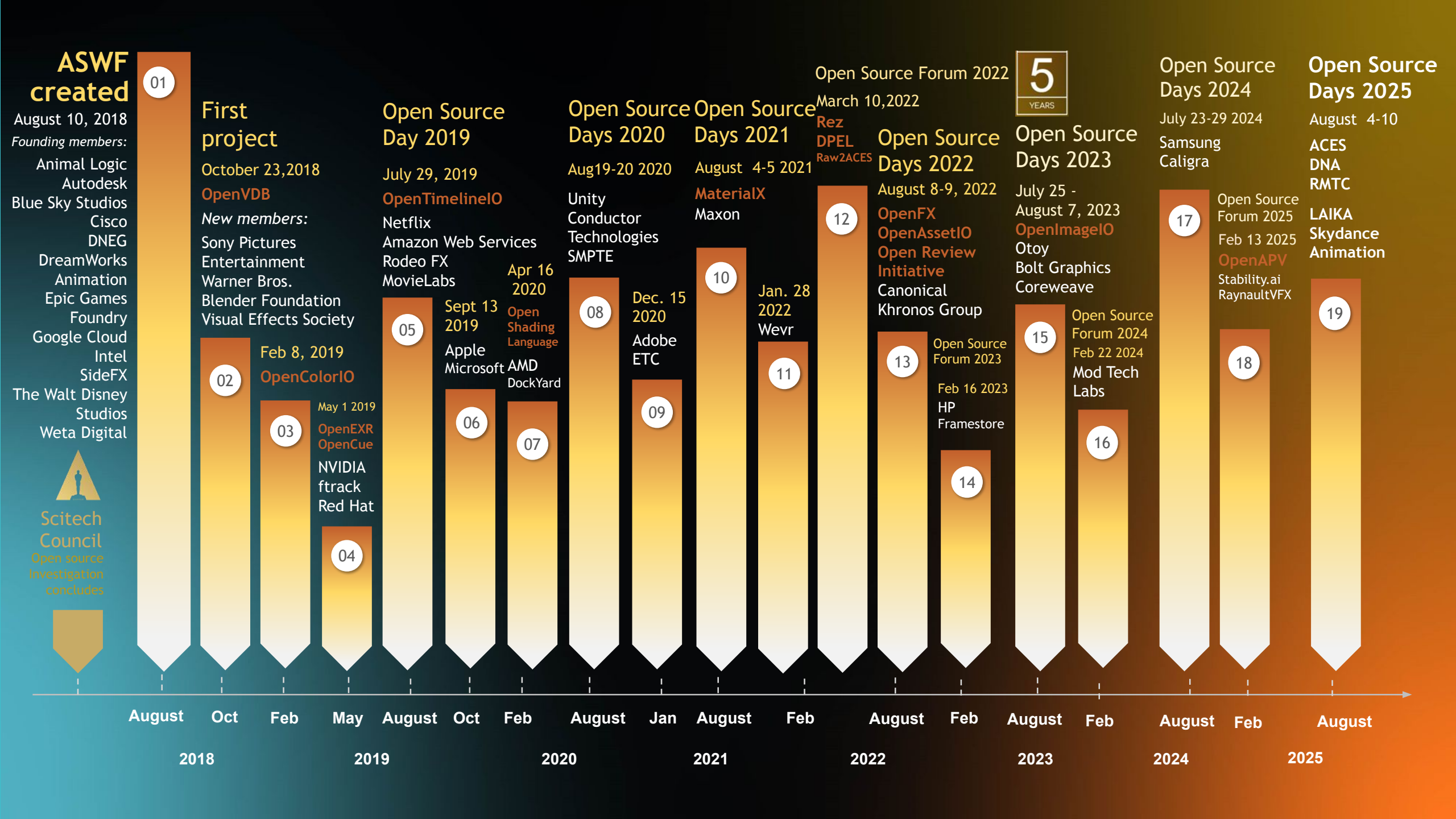
The Academy Software Foundation provides a neutral forum for open source software developers in the motion picture and broader media industries to share resources and collaborate on technologies for image creation, visual effects, animation and sound.

THE LINUX FOUNDATION

Why the Academy Software Foundation?

- Legal Framework
- Continuous Integration and Build Infrastructure
- Clear path to accepting contributions (neutral forum)
- Versionitis

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SOFTWARE
FOUNDATION


ASWF created
August 10, 2018
Founding members:
Animal Logic
Autodesk
Blue Sky Studios
Cisco
DNEG
DreamWorks Animation
Epic Games
Foundry
Google Cloud
Intel
SideFX
The Walt Disney Studios
Weta Digital
Scitech Council
Open source investigation concludes
01
First project
October 23,2018
OpenVDB
New members:
Sony Pictures Entertainment
Warner Bros.
Blender Foundation
Visual Effects Society
02
Feb 8, 2019
OpenColorIO
03
May 1 2019
OpenEXR
OpenCue
NVIDIA
ftrack
Red Hat
04
Open Source Day 2019
July 29, 2019
OpenTimelineIO
Netflix
Amazon Web Services
Rodeo FX
MovieLabs
05
Sept 13 2019
Apple
Microsoft
06
Apr 16 2020
Open Shading Language
AMD
DockYard
07
Open Source Days 2020
Aug19-20 2020
Unity
Conductor Technologies
SMPTE
08
Dec. 15 2020
Adobe
ETC
09
Open Source Days 2021
August 4-5 2021
MaterialX
Maxon
10
Jan. 28 2022
Wevr
11
Open Source Forum 2022
March 10,2022
Rez
DPEL
Raw2ACES
12
Open Source Days 2022
August 8-9, 2022
OpenFX
OpenAssetIO
Open Review Initiative
Canonical
Khronos Group
13
Open Source Forum 2023
Feb 16 2023
HP
Framestore
14
5 YEARS
Open Source Days 2023
July 25 - August 7, 2023
OpenImageIO
Otoy
Bolt Graphics
Coreweave
15
Open Source Forum 2024
Feb 22 2024
Mod Tech Labs
16
Open Source Days 2024
July 23-29 2024
Samsung
Caligra
17
Open Source Forum 2025
Feb 13 2025
OpenAPV
Stability.ai
RaynaultVFX
18
Open Source Days 2025
August 4-10
ACES
DNA
RMTC
LAIKA
Skydance Animation
19
August 2018
Oct 2018
Feb 2019
May 2019
August 2019
Oct 2019
Feb 2020
August 2020
Jan 2021
August 2021
Feb 2022
August 2022
Feb 2023
August 2023
Feb 2024
August 2024
Feb 2025
August 2025

Membership

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Premier





































General

































Associate

















Academy Software Foundation Governing Board CONFIDENTIAL

An Open Source Foundation Developed in Partnership



The Academy Software Foundation provides a neutral forum for open source software developers in the motion picture and broader media industries to share resources and collaborate on technologies for image creation, visual effects, animation and sound.

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TAC Representative



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Netflix, Inc.



Darin Grant
Treasurer
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john canning
Advanced Micro Devices (AMD)



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Paul Salvini
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Sebastien Miglio
Epic Games, Inc



Simon Crownshaw
Microsoft Corporation

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Netflix, Inc.



Cary Phillips
Industrial Light & Magic



Duncan Burbidge
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Erik Niemeyer
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john canning
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Paul DiLorenzo
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Rachel Rose
Industrial Light & Magic



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Autodesk



Youngkwon Lim
Samsung Electronics Co. Ltd.

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Chair



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Epic Games, Inc



Andy Jones
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Cary Phillips
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Industrial Light & Magic



Chris Hall
Advanced Micro Devices (AMD)



Chris Kulla
Epic Games, Inc



Christopher Moore
Skydance Animation, LLC



DANIEL GREENSTEIN
Sony Pictures Imageworks



Diego Tavares da Silva
Sony Pictures Imageworks



Eric Enderton
NVIDIA Corporation



Erik Niemeyer
Intel Corporation



Gordon Bradley
Autodesk



Greg Denton
Microsoft Corporation



Jean-Francois Panisset
TAC Representative
Visual Effects Society



Jonathan Gerber
LAIKA, LLC



Jonathan Stone
Lucasfilm



Ken Museth
NVIDIA Corporation



Kimball Thurston
Wētā FX Limited



Larry Gritz
Sony Pictures Imageworks



Matthew Low
DreamWorks Animation



Michael Min
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Michael B. Johnson
Apple Inc.



Nick Porcino
Pixar Animation Studios



Rachel Rose
Industrial Light & Magic



Rebecca Bever
Walt Disney Animation Studios



Ross Dickson
Amazon Web Services, Inc.



Scott Dyer
Academy of Motion Picture Arts and Sciences



Youngkwon Lim
Samsung Electronics Co. Ltd.

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Behind the Screens

aswf.io/behind-the-screens



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Our Mission

Increase the *quality* and *quantity* of open source contributions by establishing a *governance model*, a *legal framework* and a *community infrastructure* that lowers the barrier to entry for developing and using open source software.

Our Goals

- Provide a neutral forum to coordinate cross-project efforts, establish best practices and share resources.
- Develop an open continuous integration (CI) and build infrastructure to enable reference builds from the community.
- Streamline development for build and runtime environments through the sharing of open source build configurations, scripts and recipes.
- Provide individuals and organizations with a clear path for participation and code contribution.
- Provide better, more consistent licensing through a shared licensing template.
- Create working groups for exploration of new or larger projects.
- Focus on diversity and inclusion and raising the profile of software engineers in our industry.

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Our Projects

ACES (ACADEMY COLOR ENCODING SYSTEM)
Contributed By: Academy of Motion Picture Arts and Sciences
Chairperson(s): Scott Dyer

DNA (Dailies Notes Assistant)
Contributed By: Industrial Light & Magic (ILM)
Chairperson(s): Tommy Burnette

DPEL (Digital Production Example Library)
Contributed By: Academy Software Foundation
Chairperson(s): Matthew Low

DIVERSITY & INCLUSION WG
Contributed By: Academy Software Foundation
Chairperson(s): Carol Payne, Rachel Rose

MATERIALX
Contributed By: Lucasfilm
Chairperson(s): Jonathan Stone

Open Path Guiding Library (OpenPGL)
Contributed By:
Chairperson(s):

OPEN REVIEW INITIATIVE
Contributed By: Academy Software Foundation
Chairperson(s): Erik Strauss

open shading language
Contributed By: Sony Pictures Imageworks
Chairperson(s): Chris Kulla

OpenAssetIO
Contributed By: Samsung Electronics
Chairperson(s): Youngkwon Lim

OPEN ASSETIO
Contributed By: Foundry
Chairperson(s): Steven Shapiro

OpenColorIO
Contributed By: Sony Pictures Imageworks
Chairperson(s): Carol Payne

OpenCue
Contributed By: Sony Pictures Imageworks and Google Cloud
Chairperson(s): Diego Tavares Da Silva

OpenEXR
Contributed By: Lucasfilm
Chairperson(s): Cary Phillips

OpenFX
Contributed By: Open Effects Association
Chairperson(s): Gary Oberbrunner

OpenImageIO
Contributed By: Larry Gritz
Chairperson(s): Daniel Greenstein

OpenQMC
Contributed By: Framestore
Chairperson(s): Josh Bainbridge

OpenTimelineIO
Contributed By: Pixar Animation Studios
Chairperson(s): Eric Reinecke

OpenVDB
Contributed By: Dreamworks Animation
Chairperson(s): Ken Museth

rawtoaces
Contributed By: The Academy of Motion Picture Arts and Sciences
Chairperson(s): Anton Dukhovnikov

rez
Contributed By: Allan Johns
Chairperson(s): Jean-Christophe Morin, Stephen Mackenzie

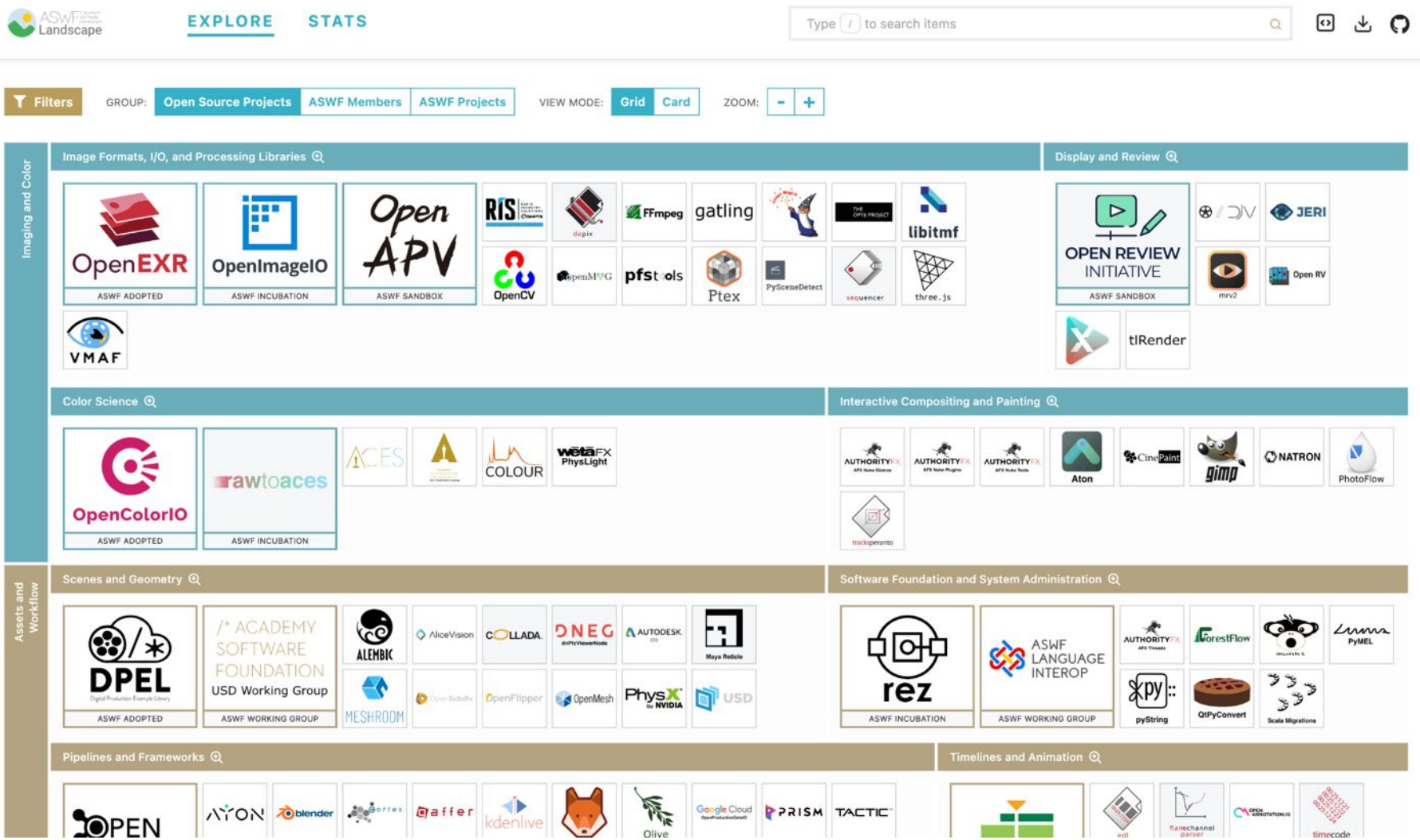
Rongota Model Train Club (RMTC)
Contributed By: Wētā FX
Chairperson(s): John Mccarten

USDWG
Contributed By: Pixar
Chairperson(s): Alexander Schwank, Nick Porcino

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ASWF Landscape



l.aswf.io

- Public list of open source projects in the animation and VFX industry
- Updates can be submitted via GitHub
- Goal is to have all active, open source projects in animation and VFX industries represented.

OPEN SOURCE FORUM

hosted by Academy Software Foundation

Annual, by-invitation event that gathers business and technical leaders to shape the future of open source software across the motion picture and media industries.
opensourceforum.aswf.io

February 26, 2026
Los Angeles, CA + Virtual

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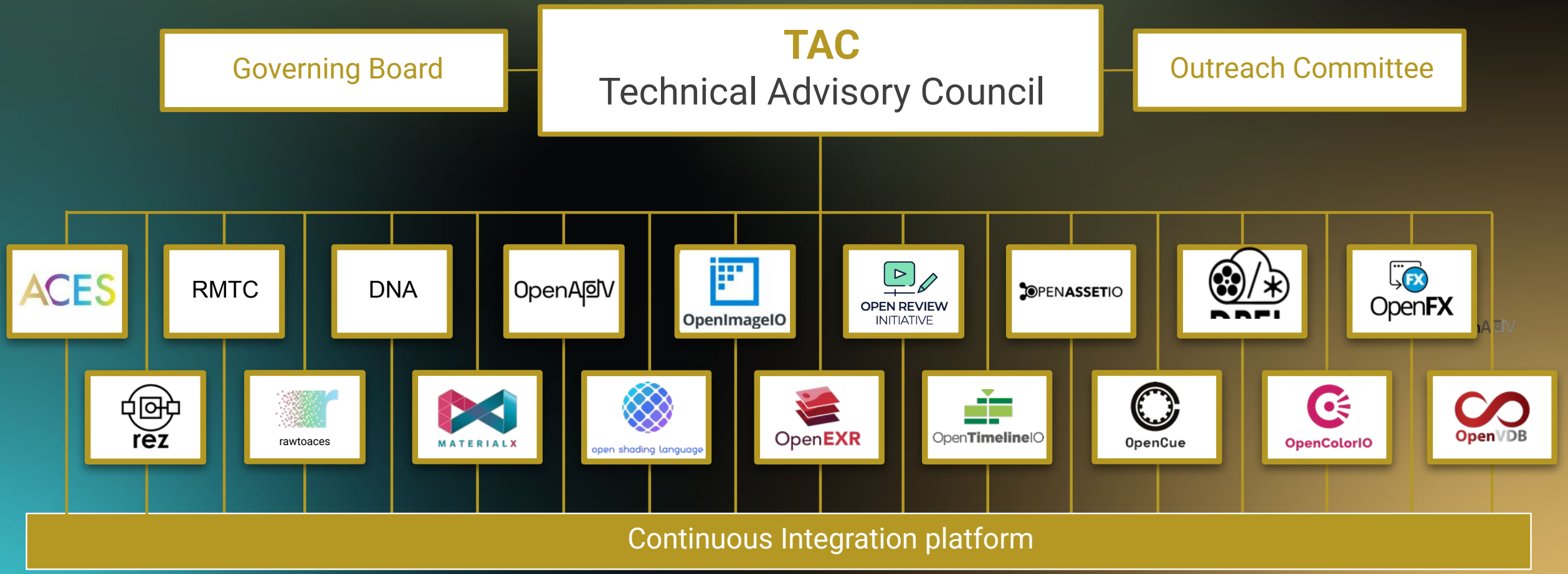
open source days '25

Annual developer focused event showcasing the latest from Academy Software Foundation projects along with other open source projects used for visual effects, animation, and image creation.
https://events.linuxfoundation.org/open-source-days/

July 19, 2026
Los Angeles, CA + Virtual

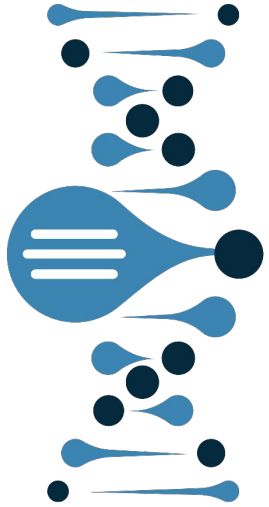
Academy Software Foundation

Structure





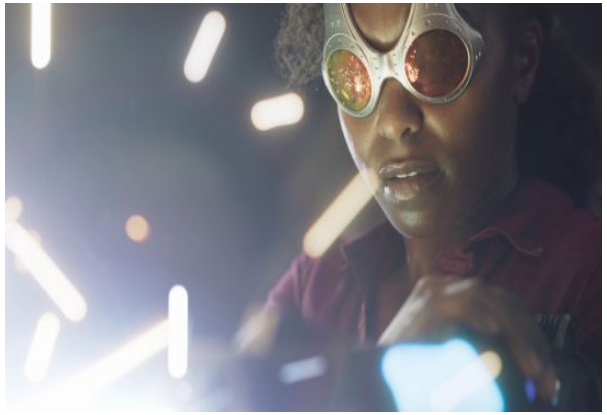
ACES (Academy Color Encoding System) is a color management and image interchange framework designed to support motion picture and television production, mastering, and archiving workflows. Developed by the Academy of Motion Picture Arts and Sciences in collaboration with industry professionals, ACES provides standardized transforms and metadata practices for consistent color fidelity across diverse tools and platforms. Originating from early digital color pipeline research in the mid-2000s, ACES has since been adopted in numerous feature films and episodic projects. Its significance lies in its role as a vendor-neutral, open solution that ensures creative intent is preserved from on-set through post and into archive, making it a cornerstone of modern professional image workflows.



DNA

Dailies Notes Assistant

The mission of the Project is to develop a tool designed to transcribe dailies meetings, analyze the content using an LLM, and provide a plugin system to integrate into production tracking software.






Launching with ASC Standard Evaluation Material II (StEM2), Animal Logic's ALab Phase 2, Intel's Volumetric Clouds Library and AWS character Noa.

DPEL

Digital Production Example Library

dpel.aswf.io

Providing a vendor-neutral hosting platform and a uniform license agreement, to encourage, curate and publish production-grade sample assets for developers, researchers and educators, Digital Production Example Library is the conclusion of an Academy Software Foundation working group on assets.

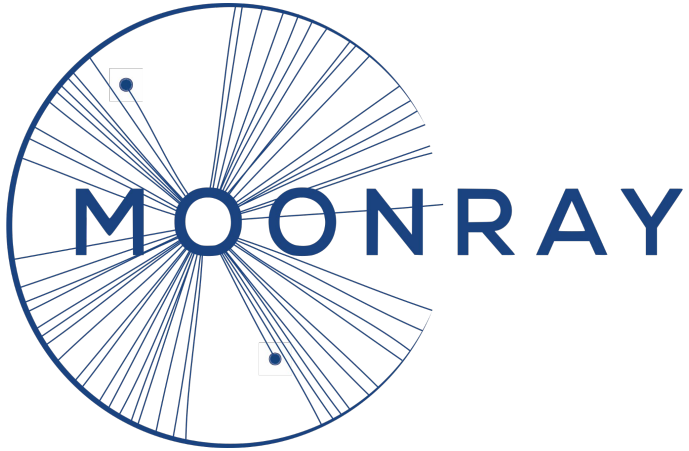


ILM Production tests using MaterialX

MATERIALX

materialx.org

- Joined ASWF in March 2021.
- Open standard for representing rich material and look-development content in computer graphics, enabling its platform-independent description and exchange across applications and renderers.
- Launched at Industrial Light & Magic in 2012, MaterialX has been a key technology in their feature films and real-time experiences since Star Wars: The Force Awakens and Millennium Falcon: Smugglers Run.
- Key contributors include Sony Pictures Imageworks, Pixar, Autodesk, Adobe, and SideFX.

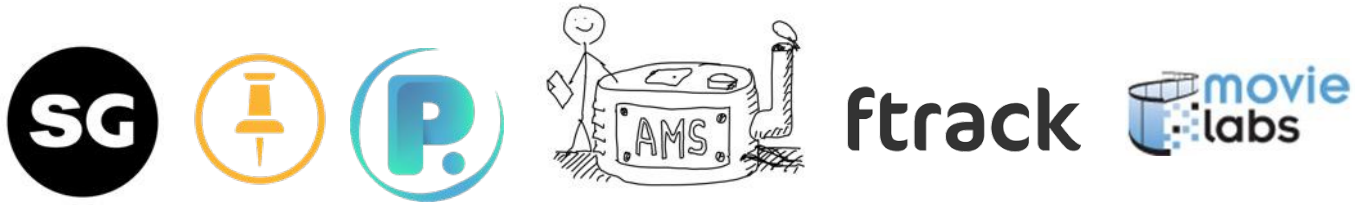


https://openmoonray.org/

OpenAPV

An open source project with the mission to develop a royalty-free, open-source, open standard video codec for professional video capturing and post-production.

- openapv-tsc@lists.aswf.io
- openapv-discussion@lists.aswf.io





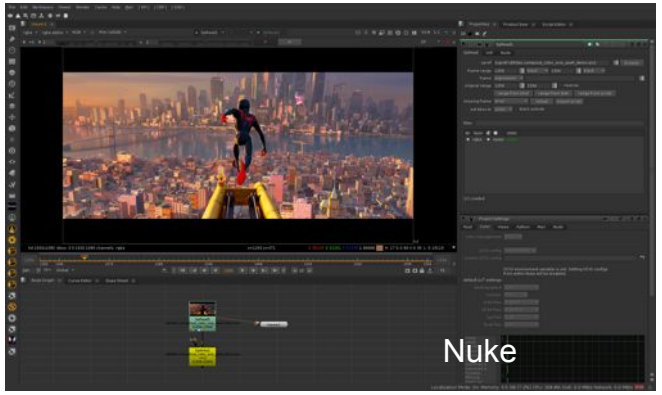
OPENASSETIO

openassetio.org





OpenAssetIO is an early stage open source interoperability standard between tools and asset management systems, developed by Foundry with proofs of concept and design input from 15 companies including DNEG, Pixar, Amazon Web Services, Animal Logic, Blizzard, and Blender. OpenAssetIO was the first project to join the Academy Software Foundation in sandbox stage.







Spider-Man™: Into the Spider-Verse; all images courtesy of Imageworks

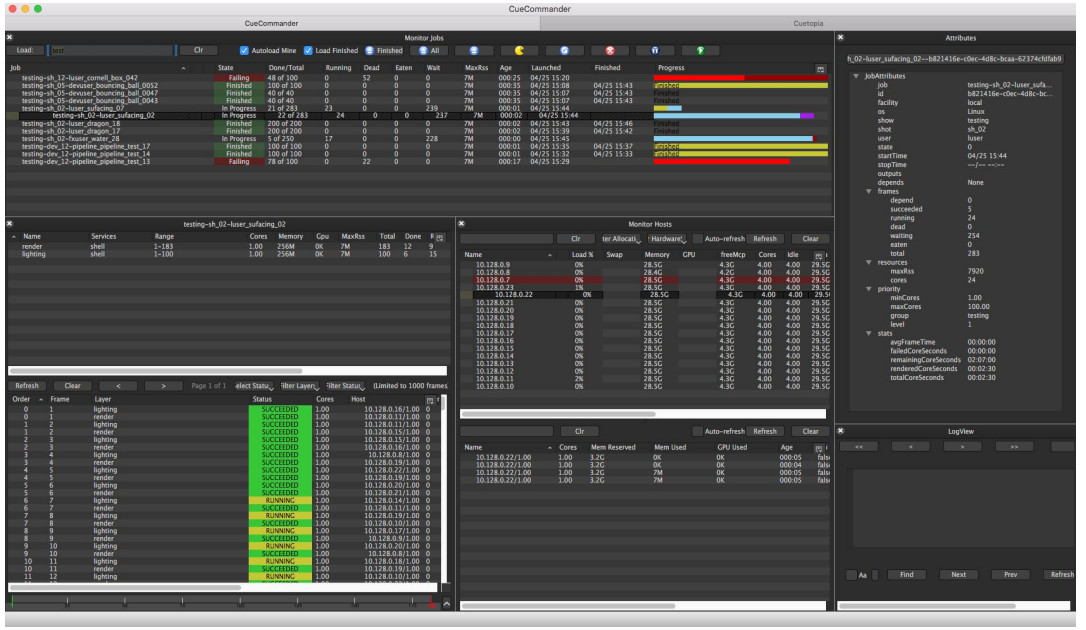
OpenColorIO

opencolorio.org

- Joined ASWF in Feb. 2019; adopted status in Nov. 2019.
- Developed by Sony Pictures Imageworks in 2003.
- Provides a consistent user experience across digital content creation applications; widely adopted in VFX and computer animation studios and software.
- Academy Technical Achievement Award recipient.
- Key contributors include Imageworks, Autodesk, DNEG, ILM, Foundry, Framestore.
- OCIO v2 released fall 2020, led by Autodesk.



Smurfs: The Lost Village; courtesy of Sony Pictures Imageworks



OpenCue screenshot

OpenCue

opencue.io

- Joined ASWF in April 2019.
- Developed at Sony Pictures Imageworks as its internal render farm manager (“Cue3”); collaboration with Google Cloud to release as OpenCue in January 2019.
- Fully featured render manager for media and entertainment that can be used to break down complex jobs into individual tasks.
- Contributors include Imageworks, Google, Netflix, Microsoft, Amazon.
- Monthly releases over the past year, including adding Python 3 compatibility, Windows support, new documentation.





Images from *Solo: A Star Wars Story.* © and TM Lucasfilm Ltd. All Rights Reserved.

OpenEXR

openexr.org

- Joined ASWF in April 2019; adopted status in Nov. 2019.
- Developed by ILM in 1999; first release of a major open source project by a studio.
- High dynamic-range (HDR) image file format for high-quality image processing and storage; widely adopted standard in computer graphics.
- Academy Technical Achievement Award recipient.
- Key contributors include ILM, Disney Animation, Imageworks, Epic Games, Weta Digital, Pixar, Autodesk, and DreamWorks.
- OpenEXR 3.0 just released.



openeffects.org

- Industry-wide common interface between image-based visual effects plugins and host applications
- Used by Autodesk Flame, Foundry Nuke, Blackmagic Design DaVinci Resolve and Fusion, Sony Catalyst and MAGIX Vegas Pro, Assimilate Scratch, Filmlight Baselight, Boris FX Sapphire and Silhouette, RE:Vision Effects and more.
- Open, extensible C API currently version 1.4 with version 1.5 in development.



Miles Morales (Shameik Moore) and Gwen Stacy (Hailee Steinfeld) take on The Spot (Jason Schwartzman) in Columbia Pictures and Sony Pictures Animation's SPIDER-MAN™: ACROSS THE SPIDER-VERSE.

OpenImageIO

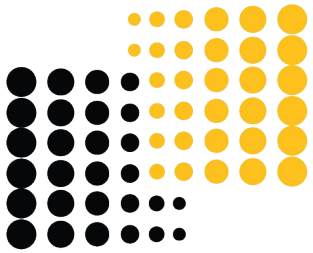
openimageio.org

OpenImageIO is a library for reading and writing images, and a bunch of related classes, utilities, and applications. There is a particular emphasis on formats and functionality used in professional, large-scale animation and visual effects work for film. OpenImageIO is used extensively in animation and VFX studios all over the world, and is also incorporated into several commercial products.

Open Path Guiding Library (OpenPGL)

Open Path Guiding Library (OpenPGL) is an open-source library, that improves the rendering efficiency of a renderer by providing easy to integrate and production-ready implementations of state-of-the-art path guiding algorithms. Especially in challenging illumination setups (complex indirect illumination, volumetric multi scattering) these algorithms significantly improve rendering performance.

openpgl.org



OpenQMC

The mission of the Project is to provide an API to deliver high quality QMC sample points.

https://openqmc.org/



DNEG's xSTUDIO



Autodesk's RV Player

OPEN REVIEW INITIATIVE

openreviewinitiative.org

An open source project with the mission to build a unified toolset for playback, review and approval of motion picture and related professional media, with Autodesk contributing RV, Dneg contributing xSTUDIO, Imageworks contributing parts of itView, and the three companies providing lead architects and maintainers for the project Technical Steering Committee.



The Angry Birds Movie 2; courtesy of Imageworks



Spider-Man: Far From Home; courtesy of Imageworks

open shading language

openshadinglanguage.org

- Joined ASWF in April 2020.
- Developed by Sony Pictures Imageworks; released as open source in 2010.
- De facto standard shading language for VFX and animation, ideal for describing materials, lights, displacement, and pattern generation.
- Academy Technical Achievement Award recipient.
- Key contributors include Imageworks, Animal Logic, NVIDIA, Pixar, DNEG, Autodesk.



OpenTimelineIO used on *Coco* for transitions and linking media to the latest renders.

Image courtesy of Pixar Animation Studios.

OpenTimelineIO

opentimeline.io

- Joined ASWF in July 2019.
- Developed by Pixar Animation Studios.
- Open source interchange format for editorial timeline information – clips, timing, tracks, transitions, metadata etc.
- Key contributors include Pixar, Netflix, DreamWorks, Autodesk.
- In Public Beta



The Croods; courtesy of DreamWorks Animation

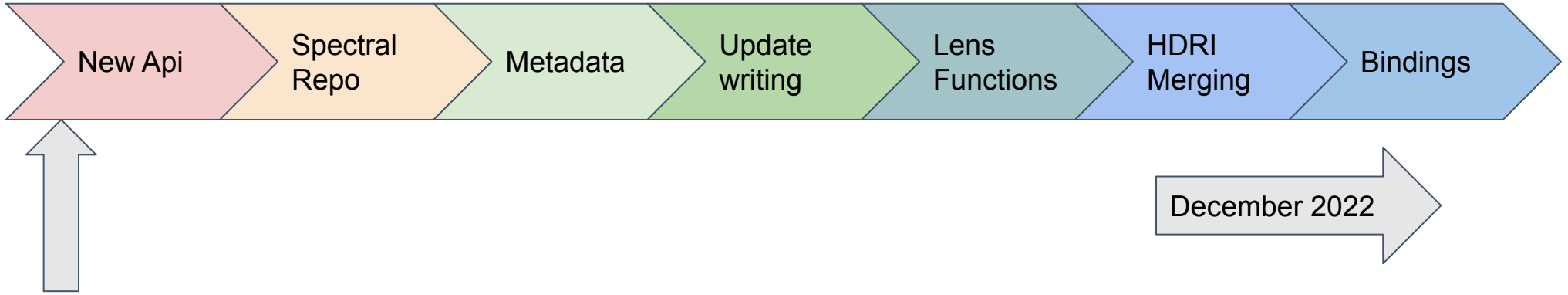


How to Train Your Dragon: The Hidden World; courtesy of DreamWorks Animation

OpenVDB

openvdb.org

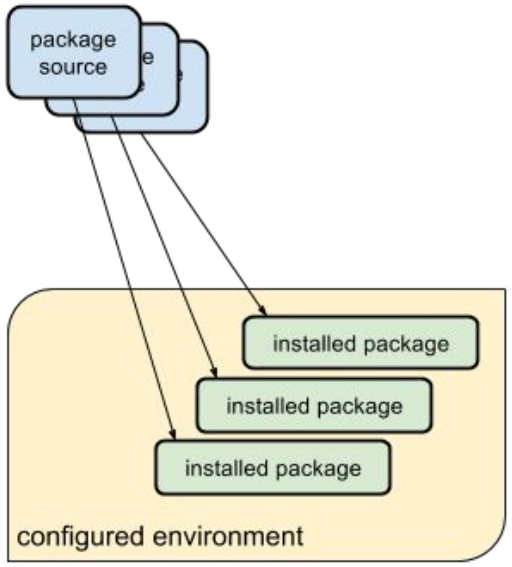
- First project adopted by ASWF in Sept. 2018; first project to graduate to adopted status in July 2019.
- Developed by DreamWorks Animation in 2012.
- C++ library for volumetric representations including fluid simulations, animated volumes and environmental effects.
- Academy Technical Achievement Award recipient.
- Key contributors include DreamWorks, Autodesk, DNEG, Industrial Light & Magic, NVIDIA, SideFX, Weta Digital.



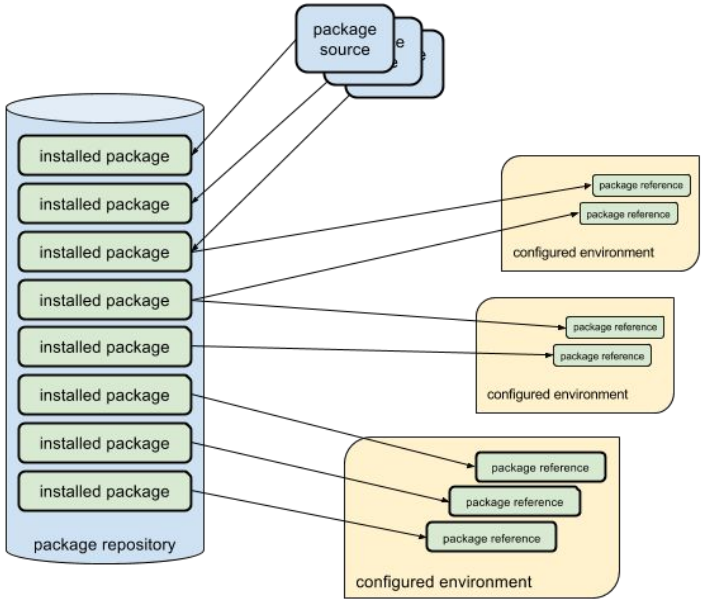
rawtoaces

rawtoaces.org

- Joined ASWF in January 2021.
- Software package that converts digital camera raw files to ACES container files containing image data encoded according to the Academy Color Encoding Specification (ACES) as specified in SMPTE 2065-1.
- Key contributors include AMPAS and Weta Digital



Typical package managers install packages into an environment



Rez installs packages once, and configures environments dynamically

rez

rez-project.io

- Joined ASWF in February 2022.
- Developed by Allan Johns of Method Studios.
- Cross-platform package manager that leverages a central repository for storing packages instead of installing into each environment separately, reducing the time to create complex environments to just a few seconds.
- Key contributors include Luma Pictures, Method Studios, NVIDIA, and Shotgrid

Working Groups

Machine Learning Working Group

A working group where the ML experts among our projects and members can meet, share expertise, better understand what ML-based tools are needed in our industry, and where helpful, spin off and guide ML-based open source projects aimed at addressing the specific needs of our film pipelines. #wg-ml

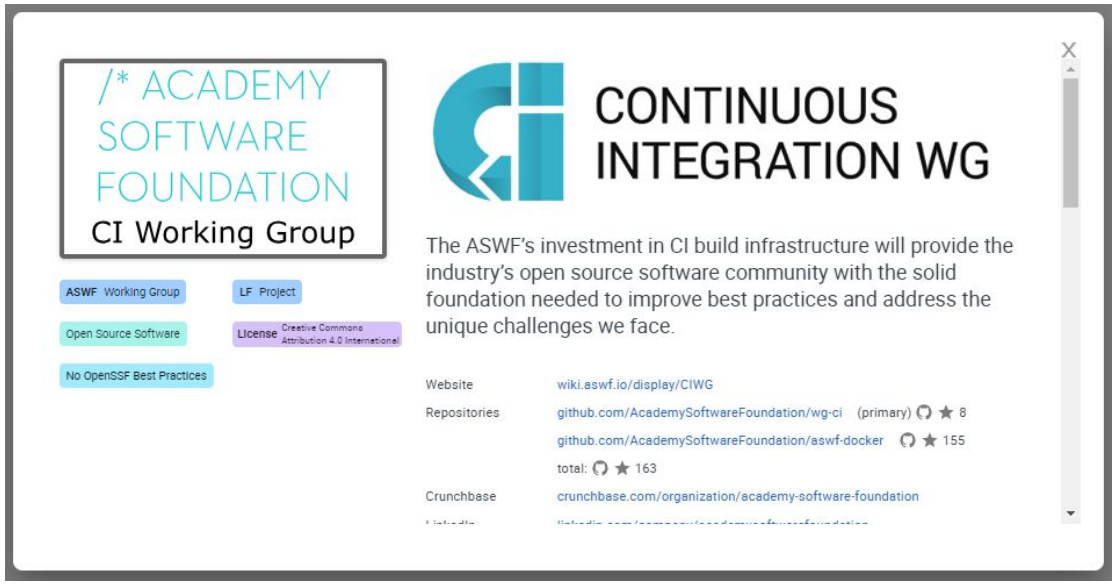
Rongotai Model Train Club (RMTC)

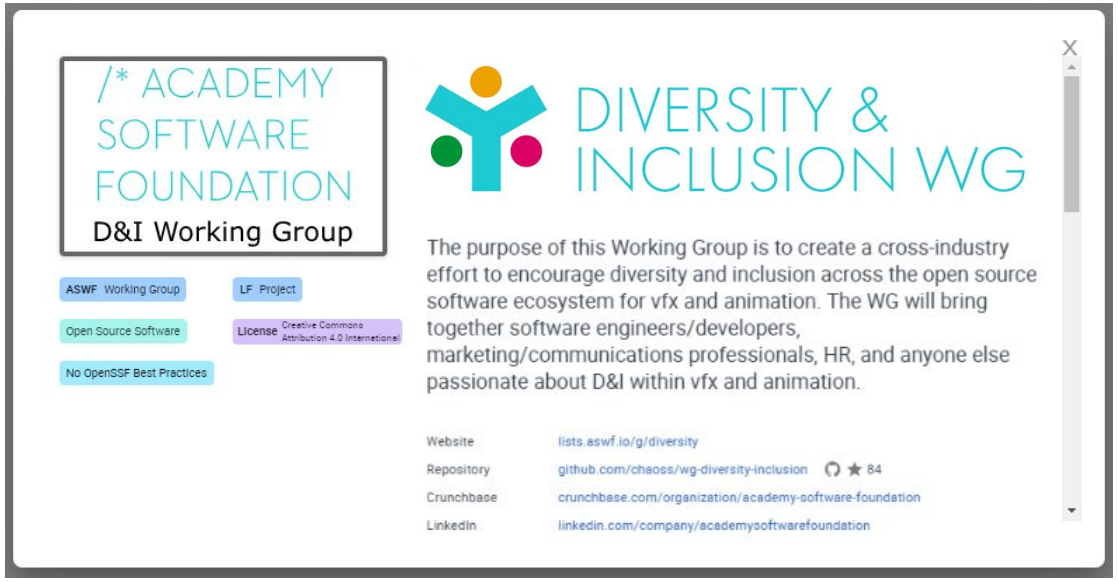
RMTC is a VFX-specific framework for simplifying the production and deployment of AI/ML models using well-tracked and rights-cleared datasets.

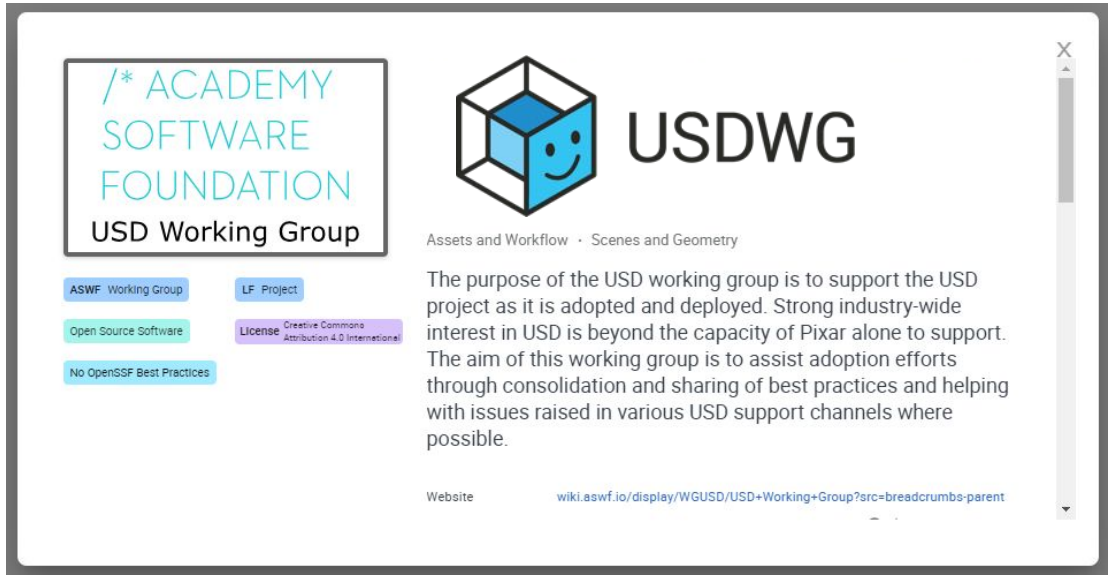
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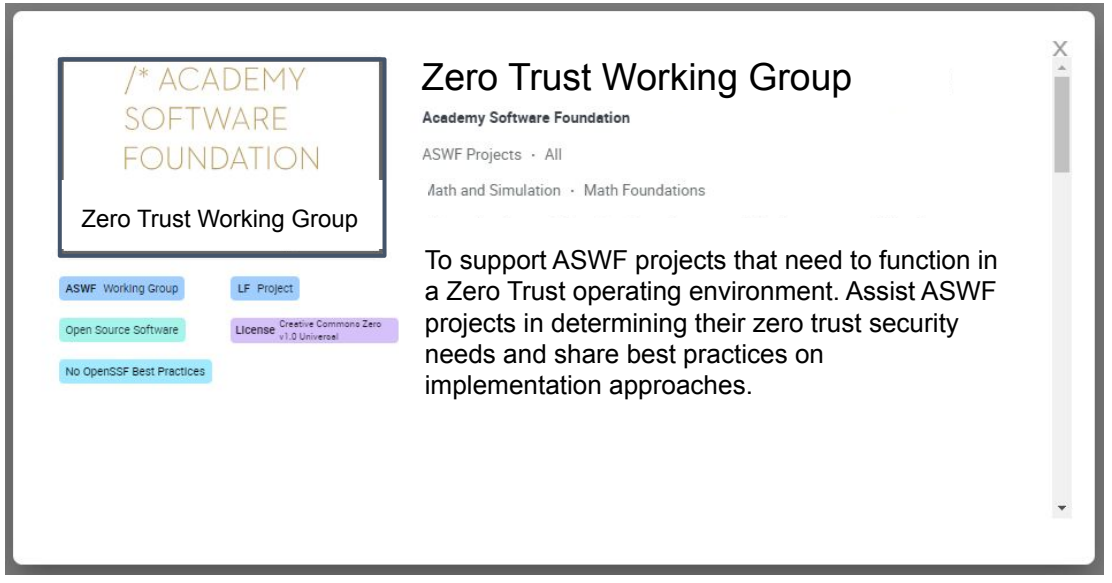
#ASWF

Working Groups









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How to contribute

Contribute to an existing project:

- Submit a PR with a bugfix or new feature
- Pick existing GitHub issue as a sample project
- Integrate with a new app / contribute plugin
- Help improve project documentation
- Submit additional test cases
- Join the TSC discussion: dev mailing list, conf calls

Get involved with the TAC or Working Group:

- DevOps / CI expertise always useful
- Interest / expertise in security especially welcome
- Join the TAC discussions: mailing list, conf calls, working groups

More information at tac.aswf.io/engagement

How to participate

Developers can join our mailing lists to stay informed about the code and our projects at lists.aswf.io.

Learn how your organization can support the Academy Software Foundation community at aswf.io/join.

Stay connected to us on Twitter at @AcademySWF and #ASWF

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THANK YOU!

Learn more at aswf.io